1.

Process	Arrival time	Burst time	Priority
P_1	0	3	4
P_2	1	6	2
P_3	4	4	5
P_4	5	5	1
P_5	6	2	3
P_6	8	4	6

(a) FCFS

(i) Without arrival time.

P_1	P_2	P_3	P_4	P_5	P_6

0 3 9 13 18 20 24

Final time	Turn around time	Waiting time
~~0~~	~~0~~	
3	3	0
9	9	3
13	13	9
18	18	13
20	20	18
24	24	20

ATAT = 14.5

AWT = 10.5

(ii) with arrival time

P_1	P_2	P_3	P_4	P_5	P_6

0 3 9 13 18 20 24

FT	TAT	WT
3	3	0
9	8	2
13	9	5
18	13	8
20	14	12
24	16	12

ATAT = 10.5

AWT = 6.5

(b) SJF

(i) Without arrival time

P_5	P_1	P_3	P_6	P_4	P_2

0 2 5 9 13 18 24

	FT	TAT	WT
P_1	5	5	2
P_2	24	24	18
P_3	9	9	5
P_4	18	18	13
P_5	2	2	0
P_6	13	13	9

ATAT = 11.83

AWT = 7.83

(b) Preemptive (with arrival time.

P_1	P_1	P_2	P_3	P_3	P_3	P_5	P_6	P_2	P_4

0 1 3 4 5 6 8 10 14 19 24

$P_1(2)$ $P_2(6)$ $P_2(5)$ $P_2(5)$ $P_2(5)$ $P_2(5)$
$P_2(6)$ $P_3(4)$ $P_3(3)$ $P_3(2)$ $P_4(5)$
$P_4(5)$ $P_4(5)$ $P_5(2)$
$P_5(2)$ $P_6(4)$

Process	FT	TAT	WT
P_1	3	3	0
P_2	19	18	12
P_3	8	4	0
P_4	24	19	14
P_5	10	4	2
P_6	14	6	2

ATAT = 9

AWT = 5

(c) Non-Preemptive (with arrival time)

P_1	P_2	P_5	P_3	P_6	P_4

0 3 9 11 15 19 24

$P_3(4)$
$P_4(5)$
$P_5(2)$
$P_6(4)$

Process	FT	TAT	WT
P_1	3	3	0
P_2	9	8	2
P_3	15	11	7
P_4	24	19	14
P_5	11	7	5
P_6	19	11	7

ATAT = 9.83
AWT = 5.83

(c) PRIORITY:

(i) without arrival time

P_4	P_2	P_5	P_1	P_3	P_6

0 5 11 13 16 20 24

Process	FT	TAT	WT
P_1	16	16	13
P_2	11	11	5
P_3	20	20	16
P_4	5	5	0
P_5	13	13	11
P_6	24	24	20

ATAT = 14.83
AWT = 10.83

(ii) Preemptive (With arrival time)

P_1	P_2	P_2	P_4	P_4	P_4	P_2	P_5	P_1	P_3	P_6

0 1 4 5 6 8 10 12 14 16 20 24

$P_1(2)-4$ $P_1(2)-4$ $P_1(2)-4$ $P_1(2)-4$ $P_1(2)-4$

$P_2(6)-2$ $P_2(3)-2$ $P_2(2)-2$ $P_2(2)-2$ $P_2(2)-2$

$P_3(4)-5$ $P_3(4)-5$ $P_3(4)-5$ $P_3(4)-5$

$P_4(5)-1$ $P_4(4)-1$ $P_4(2)-1$

$P_5(2)-3$ $P_5(2)-3$

$P_6(4)-6$

Process	FT	TAT	WT
P_1	16	16	13
P_2	12	11	5
P_3	20	16	12
P_4	10	5	0
P_5	14	8	6
P_6	24	16	12

ATAT = 12

AWT = 8

(iii) Non-Preemptive (with arrival time)

P_1	P_2	P_4	P_5	P_3	P_6

0 3 9 14 16 20 24

Process	FT	TAT	WT
P_1	3	3	0
P_2	9	8	2
P_3	20	16	12
P_4	14	9	4
P_5	16	10	8
P_6	24	16	12

ATAT = 10.33

AWT = 6.33

ROUND ROBIN (Time Slice 3ns)

P_1	P_2	P_3	P_4	P_5	P_6	P_3	P_4	P_6

0 3 6 9 12 14 17 18 20

Ready Queue

~~P_1~~	~~P_2~~	~~P_3~~	~~P_4~~	~~P_5~~	~~P_3~~	P_4	P_6
3	6	4	5	2	1	2	5

P_1	P_2	P_3	P_4	P_5	P_2	P_6	P_3	P_4	P_6

0 3 6 9 12 14 17 20 21 23 24

Ready Queue

~~P_1~~	~~P_2~~	~~P_3~~	~~P_4~~	~~P_5~~	~~P_2~~	~~P_6~~	~~P_3~~	~~P_4~~	~~P_6~~
3	6	4	5	2	3	4	1	2	1

Process	FT	TAT	WT
P_1	3	3	0
P_2	17	16	10
P_3	21	17	13
P_4	23	18	13
P_5	14	8	6
P_6	24	16	12

ATAT = 13

AWT = 9

2. GIVEN:

Person	Arrival Time	Pages (Burst time)	Priority
Worker 1	10.00	10	3
Worker 2	10.01	5	3
Manager	10.01	4	1
Worker 3	10.02	100	3
Supervisor	10.05	12	2

(1) FCFS

W1	W2	M	W3	S

10.00 10.10 10.15 10.19 11.59 12.11

Person	AFT	TAT	WT
Worker-1	10.10	10	0
Worker 2	10.15	14	9
Manager	10.19	18.	14
Worker 3	11.59	117	17
Supervisor	12.11	126	114

ATAT = 57

AWT = 30.8

2. SJF

W1	M	W2	S	W3

10 14 19 31 131

Person	FT	TAT	WT
W1	10	10	0
W2	19	18	13
W3	131	129	29
M	14	13	9
S	31	26	14

ATAT = 39.2
AWT = 13

3. SRTF

W1	M	W2	W1	S	W3

0 1 5 10 19 31 131

Person	FT	TAT	WT
W1	19	19	9
W2	10	9	4
M	5	4	0
W3	131	129	29
S	31	26	14

ATAT = 37.4
AWT = 11.2

4. Round Robin

W1	W2	M	W3	W1	W2	M	S	W3	W1	W2	S

0 2 4 6 8 10 12 14 16 18 20 21 23

Ready Queue

W3	W1	S	W3	W1	W3

26 27 28 30 32 ~~124~~

~~W1~~	~~W2~~	~~M~~	~~W3~~	~~W1~~	~~W2~~	~~M~~	~~S~~	~~W3~~	~~W1~~	~~W2~~	~~S~~	~~W3~~

10 5 4 100 8 3 2 5 98 6 1 3 96

~~W1~~	~~S~~	~~W3~~	~~W1~~	~~W3~~

4 1 94 2

Person	FT	TAT	WT
Worker 1	33	33	23
Worker 2	21	20	15
Manager	16	15	11
Worker 3	131	129	26
Supervisor	43	38	29

ATAT = 47

AWT = 20.8

5. Non-Preemptive

W1	M	S	W2	W3

0 10 14 26 31 131

Person	FT	TAT	WT
Worker 1	10	10	0
Worker 2	31	30	25
Manager	14	13	9
Worker 3	131	129	29
Supervisor	26	21	9

ATAT = 40.6

AWT = 14.4

6. Preemptive

W1	M	S	W1	W2	W3

0 1 5 17 26 31 131

Person	FT	TAT	WT
Worker 1	26	26	16
Worker 2	31	30	25
Manager	5	4	0
Worker 3	131	129	29
Supervisor	17	12	0

ATAT = 40.2

AWT = 14

3.

Process	Burst time	Priority
P_1	77	3
P_2	82	1
P_3	87	4
P_4	92	2

(a) SJF

P_1	P_2	P_3	P_4

0 77 159 246 338

Process	FT	TAT	WT
P_1	77	77	0
P_2	159	159	77
P_3	246	246	159
P_4	338	338	246

ATAT = 205

AWT = 120.5

(b) Priority Scheduling (Non-Preemptive)

P_2	P_4	P_1	P_3

0 82 174 251 338

Process	FT	TAT	WT
P_2	82	82	0
P_4	174	174	82
P_1	251	251	174
P_3	338	338	251

ATAT = 211.25

AWT = 126.75